

Programming Fundamentals – COC1071

Lab Plan – 12



Learning Objective:

- *Multidimensional Arrays*
- *Basics of Pointers*

- 1) Write a program to input a 2x3 matrix (2 rows and 3 columns) from the user and display it in matrix form.
- 2) Write a program to calculate the sum of all elements in a 3x3 matrix provided by the user.
- 3) Implement a program to find the transpose of a 2x2 matrix.
- 4) Write a program to search for a specific element in a 2D matrix and print its location (row and column).
- 5) Write a program to find the largest element in a 3x3 matrix.
- 6) Create a program to print the diagonal elements of a 3x3 matrix.
- 7) Write a program to check if a 3x3 matrix is an identity matrix.
- 8) Implement a program to add two matrices of size 3x3 entered by the user.
- 9) Write a program to multiply a 3x3 matrix by a scalar value entered by the user.
- 10) Implement a program to print the lower triangular elements of a 3x3 matrix.
- 11) Write a program to count the number of zeroes in a 5x5 matrix.
- 12) Implement a program to compare two pointers and demonstrate which one points to a higher memory address.
- 13) Write a function that takes pointers to two integers and calculates their sum and product. Use separate pointers to return the results to the calling function.
- 14) Write a function that takes pointers to two integers and swaps their digits (e.g., 23 and 45 become 32 and 54).
- 15) Create a function that returns a pointer to the larger of two integers passed to it.
- 16) Write a function that calculates the factorial of a number and stores the result in a variable passed by a pointer.